

Waiz Khan

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EDUCATION

Johns Hopkins University

M.S.E. in Data Science

• Coursework: Machine Translation, Artificial Intelligence, Deep Learning, Statistical Learning

Baltimore, MD

Aug. 2024 – May 2027

GD Goenka University

B.Tech. in Computer Science & Engineering

Gurugram, India

Aug. 2019 – Nov. 2023

RESEARCH EXPERIENCE

Research Assistant

Johns Hopkins University, Advisor: Prof. Philipp Koehn

Dec. 2025 – Present

Baltimore, MD

- Design and validate evaluation metrics for LLM representations: built an intrinsic multilinguality metric and tested it across 17 open models against 30+ preregistered predictions and held out families
- Run the experimental program on a Slurm GPU cluster, from large scale representation extraction to intervention studies across 6 training objectives and 3 seeds; author the technical reports behind an NSF proposal in development with Prof. Koehn
- Earlier phase: benchmarked 15+ decoder architectures and learning rules for hieroglyph to text translation on a 61k example corpus; best decoders reach BLEU-4 30.3, ten times the Transformer baseline

Research Assistant

GD Goenka University, Advisor: Dr. Yogesh Kumar

Aug. 2022 – Nov. 2022

Gurugram, India

- Studied chronic school absenteeism on the IHDS-II national survey of 40k+ students; logistic regression, KNN and random forest models reached 74% accuracy under fair validation
- Independently rebuilt the study in 2026 over the full 47,027 student cohort: an Optuna tuned gradient boosting ensemble reaches ROC-AUC 0.841 against a 0.699 logistic baseline, DoWhy confirms the distance barrier causally (ATE 0.14 per SD, refutation tests pass), and an integer program allocating a fixed budget doubles expected impact against random targeting

EXPERIENCE

Founding Engineer & Co-Founder

Arzaic LLC, Iris: an LLM agent platform for heart failure care

Apr. 2026 – Present

Baltimore, MD

Built through the Johns Hopkins Pava Center accelerator

- Architected a staged LLM agent pipeline (gate → route → retrieve → generate → guard) with an embedding based semantic router, agentic retrieval tools over a multimodal patient record, and a safety judge that fails closed on every response
- Built multimodal RAG that captions clinical artifacts before embedding them, over a versioned patient record that changes only through auditable patches
- Made safety evaluation a merge gate, so no change ships without passing it; a nightly agent drafts care plan revisions for clinician review

PROJECTS

Glyphos | Python, PyTorch, scikit-learn

Aug. 2026

- Evaluation infrastructure for decipherment research over eleven ancient language corpora; an audit of the standard test splits found roughly 34% near duplicate leakage, the class of error that inflates every downstream result, which motivated 26 frozen partitions and a CI gate blocking pretrained weights

ExoSAGE | Python, FastAPI, D3

Jul. 2026

- Point in time screening of NASA TOI transit candidates with false discovery rate control and walk forward backtesting, plus mixed integer telescope hour allocation under correlated risk; built on checksummed, immutable data snapshots and real TESS light curves, with 11 test suites and a research console for inspecting candidates

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, Swift, Bash

LLMs & Agents: agent pipelines and tool use, RAG and embedding retrieval, prompt engineering, agent evaluation harnesses and safety gating, Hugging Face Transformers